

A Scalable Dynamic Capacitive Wireless Power Transfer System Architecture

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Abstract—This paper presents a novel power-scalable architecture for dynamic capacitive wireless power transfer (WPT) systems utilizing an active variable reactance (AVR) rectifier. The AVR rectifier enables continuous compensation, allowing the WPT system to maintain high-efficiency power transfer at a fixed frequency by appropriately distributing power between its two branches. A transformer-based power-splitting method is used to realize a scalable AVR rectifier while maintaining high system efficiency and wide dynamic compensation range. Furthermore, a state-space model of this architecture is developed and used to design a full state feedback controller. To validate the proposed approach, a 6.78 MHz 12-kW dynamic capacitive WPT system is built. Three 4-kW full-bridge inverters and three half-bridge rectifiers per AVR branch are paralleled to achieve this power level. The system's operation is demonstrated through both simulations and experiments on the hardware prototype.

I. INTRODUCTION

Wireless power transfer (WPT) is becoming increasingly important and has the potential to accelerate the widespread adoption of electric vehicles (EVs), especially amongst vehicles that routinely travel along fixed routes such as buses and warehouse forklifts [1], [2]. Capacitive WPT systems can be lighter, cheaper, and smaller than comparable inductive WPT systems due to the absence of ferrite cores [3]. The ability of capacitive WPT systems to transfer high power at high efficiency has been demonstrated, thus proving the viability of this technology [4].

One main challenge of WPT systems is the extreme sensitivity to the value of the coupling capacitance or mutual inductance between the receiver and the transmitter [5]. To increase the practicality of WPT systems in dynamic applications, it is desirable to design and control them to tolerate coupling variations while maintaining full power transfer at high efficiency [6]–[9]. The recently proposed active variable reactance (AVR) rectifier addresses these challenges and efficiently provides variable compensation in high-frequency WPT systems [10]. Additionally, a novel time-domain solution for the control of the AVR rectifier has been proposed in [11] and demonstrated on hardware at low power. This approach allows an AVR-equipped capacitive

WPT system to dynamically compensate for changes in the coupling capacitance automatically.

Capacitive WPT systems typically operate at multi-MHz frequencies, which pose another significant challenge for high-power operation due to limitations in commercially available GaN devices. Various power-combining approaches have been proposed in [4], [12], with the transformer-based power-combining and power-splitting approach in [4] being particularly effective due to its scalability, low component count, and simple implementation. This digest combines the AVR rectifier architecture [10] with the transformer-based power-splitting method [4] to obtain a scalable dynamic capacitive WPT system.

In this work, an AVR-equipped scalable dynamic capacitive WPT system architecture is proposed. A state-space model is developed for this architecture, and a full-state-feedback controller is designed using the proposed model to automatically compensate for variations in coupling capacitance. Finally, the proposed architecture and controller are validated first in simulation, then on a prototype 6.78 MHz, 12-kW dynamic capacitive WPT hardware system.

II. DYNAMIC CAPACITIVE WIRELESS POWER TRANSFER SYSTEM WITH AN AVR RECTIFIER

Figure 1 presents the practical circuit schematic of the dynamic capacitive WPT system with the AVR rectifier. A DC input bus is connected to a full-bridge inverter. C_s is the coupling capacitance between the roadway-side and vehicle-side coupling plates. The L-section matching network (L_{pri} , C_{pri} , L_{sec} , C_{sec}) surrounding C_s is designed to provide a voltage gain to minimize displacement current through the air gap while compensating the reactance of C_s such that the loading on the inverter is near-resistive, allowing the inverter to achieve soft-switching.

The AVR rectifier has two branches: a top branch containing a boost converter attached to an inductive reactance $+jX$ through one half-bridge rectifier, and a bottom branch containing a buck converter attached to a capacitive reactance $-jX$ through another half-bridge rectifier. By independently varying the amount of power processed in each branch, the AVR rectifier can maintain the desired constant output

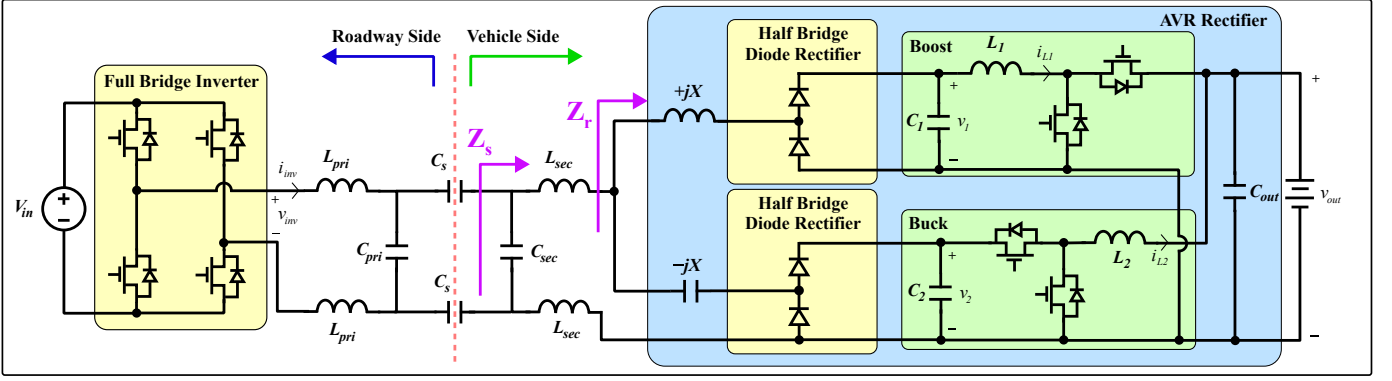


Fig. 1: Dynamic capacitive wireless power transfer system with active variable reactance rectifier schematic.

$$p_{out} = p_1 + p_2 = v_1 i_{L1} + v_{out} i_{L2} \quad (1a)$$

$$z_r = \text{Re}(z_r) + j \cdot \text{Im}(z_r) = r_r + jx_r = \frac{k_{rec}^2 v_1^2 v_2^2 + v_1 v_{out} i_{L1} i_{L2} X^2}{k_{rec} (v_1 v_2^2 i_{L1} + v_{out} v_1^2 i_{L2})} + jX \frac{v_1 i_{L1} v_2^2 - v_{out} i_{L2} v_1^2}{v_1 i_{L1} v_2^2 + v_{out} i_{L2} v_1^2} \quad (1b)$$

$$z_s = \text{Re}(z_s) + j \cdot \text{Im}(z_s) = r_s + jx_s = \frac{r_r}{(1 - \omega_s^2 C_{sec} L_{sec} - \omega_s C_{sec} x_r)^2 + (\omega_s C_{sec} r_r)^2} + j \frac{(\omega_s L_{sec} + x_r)(1 - \omega_s^2 C_{sec} L_{sec} - \omega_s C_{sec} x_r) - r_r^2 \omega_s C_{sec}}{(1 - \omega_s^2 C_{sec} L_{sec} - \omega_s C_{sec} x_r)^2 + (\omega_s C_{sec} r_r)^2} \quad (1c)$$

power (P_{out}) while compensating for coupling variations by modulating the impedance seen at the input of the AVR rectifier (z_r). Throughout this paper, the following notation is used: p_{out} is the time-varying measured variable, P_{out} is the constant nominal DC value, and $\hat{p}_{out}$ is the small-signal variable. They are related as $p_{out}(t) \equiv P_{out} + \hat{p}_{out}(t)$.

Equations 1a-1c give p_{out} , z_r , and z_s as a function of the large-signal circuit states as labeled in Figure 1. k_{rec} is a gain associated with the half-bridge rectifiers under sinusoidal approximation, X is the differential reactance $\pm jX$ attached to the two branches, and ω_s is the inverter switching frequency. When the coupling capacitance C_s changes by an amount ΔC_s , Eqns. 1a-1c are utilized to adjust the operation of the AVR rectifier such that the real part of the impedance looking into the secondary $\text{Re}(z_s)$ is held constant at its nominal value $r_s = R_s$. This also guarantees $p_{out} = P_{out}$, as well as resistive loading on the inverter. A derivation of Eqn. 1b can be found in [10]; a derivation of Eqn. 1c is provided in Appendix A.

III. SCALABLE ARCHITECTURE FOR DYNAMIC CAPACITIVE WPT

This section motivates and presents a novel topology for scaling the capacitive WPT system described in Section II to higher powers while maintaining dynamic compensation capability. In high-power capacitive WPT systems, the performance bottleneck is typically at the high-frequency inverters, as the GaN FETs used have much lower power-handling capabilities compared to the SiC Schottky diodes and switches in the rectifiers and DC-DC converters. To address inverter power scaling issues, multiple inverters can be paralleled using transformers as described in [4]. But,

the performance bottleneck of the system is now shifted to the SiC diodes in the high frequency half-bridge rectifiers in the AVR rectifier. Additionally, there is an inherent tradeoff between the efficiency of the matching network, the gain of the matching network, and the AVR rectifier's compensation range. Thus, having a large current gain while achieving a target matching network efficiency causes the maximum compensation range of the resulting AVR rectifier to be very narrow, decreasing the utility of the system. [10] To meet the growing power demands of EV chargers without compromising the dynamic compensation range of WPT systems, there is a need to develop power-scaling methods that enable effective parallel operation of the AVR rectifier.

Figure 2 shows the proposed novel scalable dynamic capacitive WPT topology. On the roadway side, an N -level stacked-inverter architecture introduced in [4] combines the power from N high-frequency inverters at the input. On the vehicle side, power is split between the $+jX$ and $-jX$ branches. On each branch, an M -level stacked-rectifier structure using a series-in parallel-out transformer splits the power into M high-frequency half-bridge diode rectifiers. The rectified power is then combined in parallel and fed through a single DC-DC converter on each branch. This topology meshes the wide dynamic compensation range of the AVR rectifier with the high-efficiency scalable system proposed in [4], all with a low component count. Moreover, the design of the matching network and AVR differential impedance X can be co-designed in the same manner as described in [10] by simply replacing V_{in} and V_{out} with an effective $V_{in,eff} = \kappa N V_{in}$ and $V_{out,eff} = \kappa M V_{out}$, where κ is the coupling coefficient of the transformers.

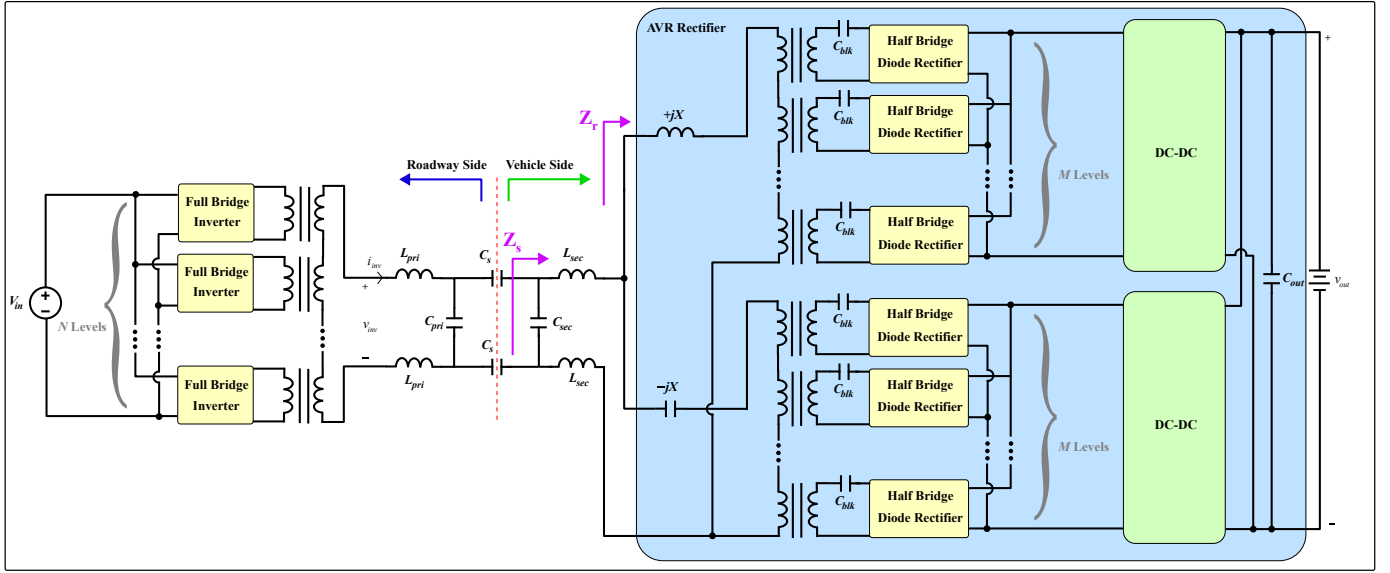


Fig. 2: Proposed scalable dynamic capacitive wireless power transfer system schematic. All transformers have a 1:1 turns ratio and a coupling factor of κ (not shown).

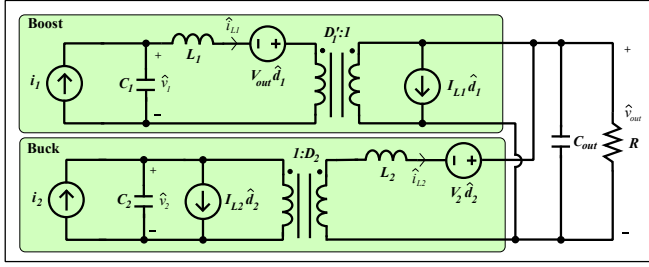


Fig. 3: Schematic of the linearized small-signal model of the AVR rectifier.

This architecture thus effectively addresses the lower power-handling capabilities of the high-frequency inverters and rectifiers by paralleling them using transformers. This architecture simultaneously minimizes component count and overall system complexity by performing the power combining at the output in front of the pair of DC-DC converters, which avoids duplicating the paired DC-DC converter structure M times. This is possible because the DC-DC converters are relatively simple and low-frequency buck and boost converters, which can be made to handle the full 12 kW of output power. The novel architecture constitutes the first significant contribution of this paper.

IV. MODELING, CONTROL, AND SIMULATION OF THE SCALABLE DYNAMIC CAPACITIVE WPT SYSTEM

In this section, a model for the scalable WPT architecture is developed in state space using methods presented in [11], [13]. A full state feedback controller is then designed utilizing this model to provide dynamic compensation for a changing coupling capacitance. Finally, the proposed topology and controller are validated in simulation. Since the proposed architecture of Fig. 2 does not modify the small-signal schematic of the AVR rectifier (Fig. 3), the derivation of the state-space model presented in [11] can be followed in much the same manner. The state vector $\hat{\vec{x}}$ contains the

small-signal voltages and currents $\hat{v}_1, \hat{v}_2, \hat{i}_{L1}, \hat{i}_{L2}$, and $\hat{v}_{out}$, as labeled in Fig. 3.

As in the matching network optimization, the values of V_{in} and V_{out} must be replaced with the effective $V_{in,eff} = \kappa N V_{in}$ and $V_{out,eff} = \kappa M V_{out}$, respectively. As this system uses an L-section matching network instead of the series-L compensation used in [11], the controller now must drive the real part of the secondary-side impedance r_s to the nominal value R_s . Therefore, the augmented state-space matrix contains the partial derivatives of r_s , taken using Eqn. 1c. The resulting state-space model is given by Eqn. 2; a derivation of the partial derivatives included in Eqn. 2 is presented in Appendix B.

Next, a full state feedback controller is designed using methods employed in [11], [14]. Figure 5 shows the complete block diagram of the system under closed-loop control. The controller design consists of finding a gain matrix K such that setting the input as $\vec{u} = \vec{D} - K\vec{x}_{aug}$ results in the closed-loop system driving p_{out} and r_s to their nominal values P_{out} and R_s regardless of the value of C_s . To find an optimal K , heuristic techniques presented in [14] are used to design a linear quadratic regulator (LQR) with reasonable values for the diagonal elements of the Q and R cost-defining matrices. Then, the `lqr()` function in MATLAB is utilized to solve the corresponding Algebraic Riccati Equation for the optimal gain matrix K .

A model is built in Simulink and PLECS to validate the proposed topology and controller for a 6.78 MHz, 12-kW system with $V_{in} = V_{out} = 400$ V. Figure 4 shows steady-state inverter output waveforms and AVR branch currents. It also shows the control objectives p_{out} and r_s being controlled to $P_{out} = 12$ kW and $R_s = 43.2 \Omega$, while the coupling reactance is disturbed by an amount $\Delta X_s \equiv \frac{1}{2\pi f_s} \left(\frac{1}{C_{s,nom}} - \frac{1}{C_s} \right)$, where f_s is the switching frequency, and $C_{s,nom}$ is the nominal coupling capacitance. The state-space model, full state feedback controller, and

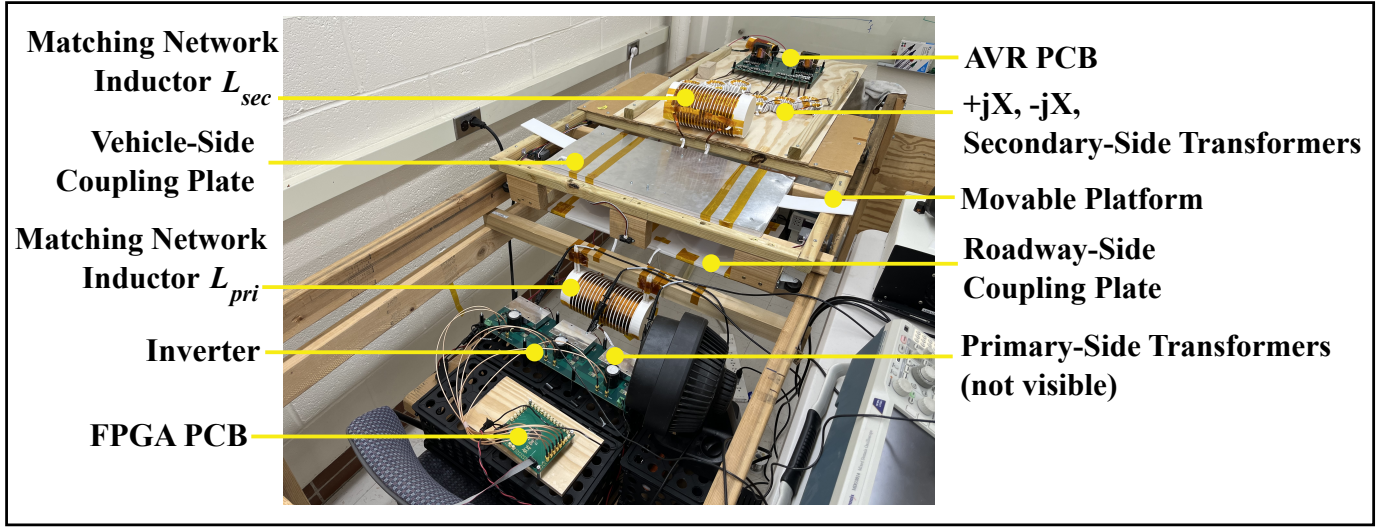


Fig. 6: Hardware setup annotated.

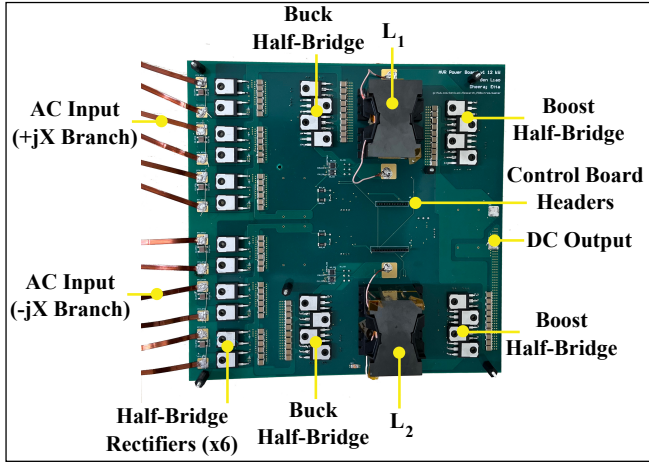


Fig. 7: Annotated power board PCB.

Note that the inductive reactance $+jX$ is not realized by adding an explicit inductor; rather, it is instead realized by adding a relatively large capacitor in series with the leakage inductance of the transformers. The hardware and experimental validation of the proposed topology are the third significant contribution of this paper.

VI. CONCLUSION AND FUTURE WORK

This paper presents a novel architecture for scalable dynamic capacitive WPT systems. A state-space model is developed and a full state feedback controller is designed for the proposed topology. The operation of the system is validated in simulation. Finally, a 6.78-MHz, 12-kW AVR-equipped scalable capacitive WPT system with three 4-kW-rated full bridge inverters and three half-bridge rectifiers in each AVR branch is built. Its operation is then validated on this hardware. The next goal will be to show the system operating with fully dynamic closed-loop control at kilowatt-

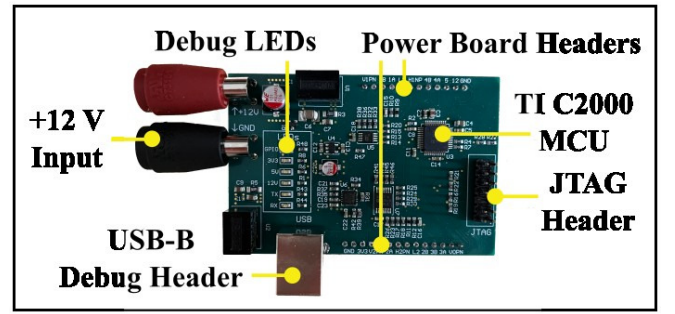


Fig. 8: Annotated control board PCB.

scale, and to validate the proposed topology on hardware up to 12 kW.

APPENDIX A

This appendix derives the expression for the impedance seen by the coupling capacitances, which is the same as the input impedance of the AVR rectifier transformer through the secondary side of the matching network. Consider Figure 10 where the AVR rectifier input impedance is given by Eqn. 1b. Then:

$$z_s = \frac{1}{j\omega_s C_{sec}} \parallel (j\omega_s L_{sec} + r_r + jx_r) \quad (3)$$

Expanding, this becomes:

$$z_s = \frac{r_r + j(\omega_s L_{sec} + x_r)}{1 - \omega_s^2 C_{sec} L_{sec} - \omega_s C_{sec} x_r + j\omega_s C_{sec} r_r} \quad (4)$$

Separating this expression into real and imaginary parts, then simplifying, gives z_s in the form $z_s = \text{Re}(z_s) + j \cdot \text{Im}(z_s) = r_s + jx_s$, where:

$$r_s = \frac{r_r}{(1 - \omega_s^2 C_{sec} L_{sec} - \omega_s C_{sec} x_r)^2 + (\omega_s C_{sec} r_r)^2}$$

$$x_s = \frac{(\omega_s L_{sec} + x_r)(1 - \omega_s^2 C_{sec} L_{sec} - \omega_s C_{sec} x_r) - r_r^2 \omega_s C_{sec}}{(1 - \omega_s^2 C_{sec} L_{sec} - \omega_s C_{sec} x_r)^2 + (\omega_s C_{sec} r_r)^2} \quad (5)$$

This is the same as Eqn. 1c.

APPENDIX B

This appendix derives the partial derivatives of p_{out} and r_s with respect to the elements of the state vector $\vec{x} = [v_1 \ v_2 \ v_{out} \ i_{L1} \ i_{L2}]^T$. p_{out} is the output power, and r_s is the real part of the impedance presented by the secondary-side matching network to the coupling capacitance.

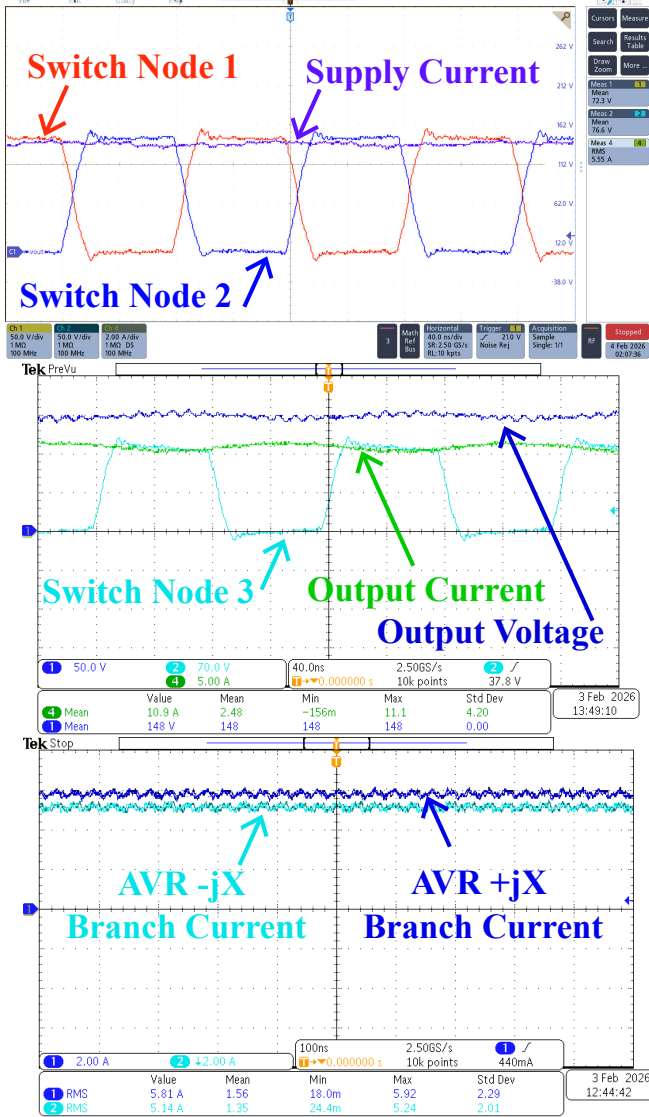


Fig. 9: Annotated system waveforms at 2 kW output power, nominal misalignment condition.

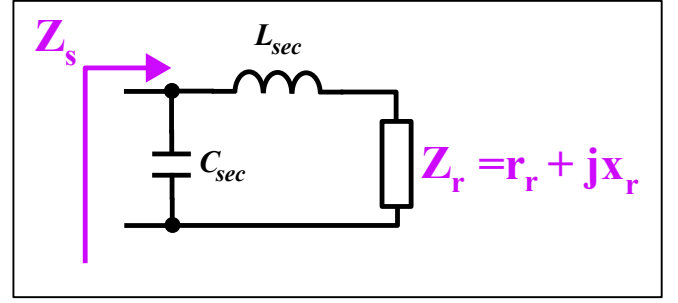


Fig. 10: Schematic illustrating z_s equivalent circuit.

The partial derivatives of p_{out} are straightforward; take them directly from Eqn. 1a to obtain:

$$\partial p_{out} / \partial \vec{x} = [I_{L2} \ I_{L1} \ 0 \ V_1 \ V_{out}]^T \quad (6)$$

The partial derivatives of r_s are much more involved. Start with the expression for r_s , given in Eqn. 1c:

$$r_s = \frac{r_r}{(1 - \omega_s^2 C_{sec} L_{sec} - \omega_s C_{sec} x_r)^2 + (\omega_s C_{sec} r_r)^2} \quad (7)$$

Expanding, this becomes:

$$r_s = \frac{r_r}{a + b x_r + c(x_r^2 + r_r^2)} \quad (8a)$$

$$a \equiv 1 - 2\omega_s^2 C_{sec} L_{sec} + \omega_s^4 C_{sec}^2 L_{sec}^2 \quad (8b)$$

$$b \equiv -2\omega_s C_{sec} + 2\omega_s^3 C_{sec}^2 L_{sec} \quad (8c)$$

$$c \equiv \omega_s^2 C_{sec}^2 \quad (8d)$$

Importantly, a , b , and c are constants with respect to the variables of interest v_1 , v_2 , v_{out} , i_{L1} , and i_{L2} . Next, recall the expressions for r_r and x_r given in Eqn. 1b:

$$r_r = \frac{k_{rec}^2 v_1^2 v_2^2 + v_1 v_{out} i_{L1} i_{L2} X^2}{k_{rec} (v_1 v_2^2 i_{L1} + v_{out} v_1^2 i_{L2})} \quad (9)$$

$$x_r = X \frac{v_1 i_{L1} v_2^2 - v_{out} i_{L2} v_1^2}{v_1 i_{L1} v_2^2 + v_{out} i_{L2} v_1^2}$$

Recall that the state vector $\vec{x}$ contains the voltages and currents v_1 , v_2 , i_{L1} , i_{L2} , and v_{out} , as labeled in Fig. 3. Define the following quantities:

$$\delta(\vec{x}) \equiv v_1 v_2^2 i_{L1} + v_{out} v_1^2 i_{L2}$$

$$\eta_r(\vec{x}) \equiv k_{rec}^2 v_1^2 v_2^2 + v_1 v_{out} i_{L1} i_{L2} X^2 \quad (10)$$

$$\eta_x(\vec{x}) \equiv v_1 i_{L1} v_2^2 - v_{out} i_{L2} v_1^2$$

Using these quantities, the expression for r_s can be written as:

$$r_s = \frac{\frac{\eta_r}{k_{rec} \delta}}{a + b \frac{\eta_x}{\delta} + c \left(\left(\frac{\eta_x}{\delta} \right)^2 + \left(\frac{\eta_r}{k_{rec} \delta} \right)^2 \right)} \quad (11a)$$

$$= \frac{k_{rec} \delta \eta_r}{a k_{rec}^2 \delta^2 + b k_{rec}^2 \delta \eta_x + c k_{rec}^2 \eta_x^2 + c \eta_r^2} \quad (11b)$$

Define the numerator and denominator of Eqn. 11b as:

$$\begin{aligned} n(\vec{x}) &\equiv k_{rec}\delta\eta_r \\ d(\vec{x}) &\equiv ak_{rec}^2\delta^2 + bk_{rec}^2\delta\eta_x + ck_{rec}^2\eta_x^2 + c\eta_r^2 \end{aligned} \quad (12)$$

Therefore, $r_s = n/d$. The desired vector of partial derivatives $\partial r_s / \partial \vec{x}$ can be written as:

$$\frac{\partial r_s}{\partial \vec{x}} = \frac{d \cdot \partial n / \partial \vec{x} - n \cdot \partial d / \partial \vec{x}}{d^2} \quad (13)$$

where:

$$\frac{\partial n}{\partial \vec{x}} = k_{rec} \left(\frac{\partial \delta}{\partial \vec{x}} \eta_r + \frac{\partial \eta_r}{\partial \vec{x}} \delta \right) \quad (14a)$$

$$\begin{aligned} \frac{\partial d}{\partial \vec{x}} &= ak_{rec}^2 \cdot 2\delta \cdot \frac{\partial \delta}{\partial \vec{x}} \\ &\quad + bk_{rec}^2 \left(\frac{\partial \delta}{\partial \vec{x}} \eta_x + \frac{\partial \eta_x}{\partial \vec{x}} \delta \right) \\ &\quad + ck_{rec}^2 \cdot 2\eta_x \cdot \frac{\partial \eta_x}{\partial \vec{x}} + c \cdot 2\eta_r \cdot \frac{\partial \eta_r}{\partial \vec{x}} \end{aligned} \quad (14b)$$

Finally, each of the partial derivatives of δ , η_r and η_x are computed and presented below:

$$\frac{\partial \delta}{\partial \vec{x}} = \begin{bmatrix} \partial \delta / \partial v_1 \\ \partial \delta / \partial v_2 \\ \partial \delta / \partial v_{out} \\ \partial \delta / \partial i_{L1} \\ \partial \delta / \partial i_{L2} \end{bmatrix} = \begin{bmatrix} V_2^2 I_{L1} + 2V_1 V_{out} I_{L2} \\ 2V_1 V_2 I_{L1} \\ V_1^2 I_{L2} \\ V_1 V_2^2 \\ V_{out} V_1^2 \end{bmatrix} \quad (15a)$$

$$\frac{\partial \eta_r}{\partial \vec{x}} = \begin{bmatrix} \partial \eta_r / \partial v_1 \\ \partial \eta_r / \partial v_2 \\ \partial \eta_r / \partial v_{out} \\ \partial \eta_r / \partial i_{L1} \\ \partial \eta_r / \partial i_{L2} \end{bmatrix} = \begin{bmatrix} 2k_{rec}^2 V_1 V_2^2 + V_{out} I_{L1} I_{L2} X^2 \\ 2k_{rec}^2 V_1^2 V_2 \\ V_1 I_{L1} I_{L2} X^2 \\ V_1 V_{out} I_{L2} X^2 \\ V_1 V_{out} I_{L1} X^2 \end{bmatrix} \quad (15b)$$

$$\frac{\partial \eta_x}{\partial \vec{x}} = \begin{bmatrix} \partial \eta_x / \partial v_1 \\ \partial \eta_x / \partial v_2 \\ \partial \eta_x / \partial v_{out} \\ \partial \eta_x / \partial i_{L1} \\ \partial \eta_x / \partial i_{L2} \end{bmatrix} = \begin{bmatrix} I_{L1} V_2^2 - 2V_1 V_{out} I_{L2} \\ 2V_1 V_2 I_{L1} \\ -I_{L2} V_1^2 \\ V_1 V_2^2 \\ -V_{out} V_1^2 \end{bmatrix} \quad (15c)$$

Plugging all of the above expressions back into Eqn. 14a and 14b, and those expressions back into Eqn. 13, along with the definitions of a , b , c , δ , η_r , η_x , n , and d , will yield the desired partial derivatives.

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